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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Final Report for
IoT Connected SmartHub dashboard

ECE436 Internet of Things

By

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Executive Summary

The IoT Connected SmartHub Dashboard is a multi-functional smart desk system developed using ESP32 microcontrollers with LoRa, Bluetooth Low Energy (BLE), and Wi-Fi communication technologies [1], [2]. The system provides real-time information including weather updates, music playback, battery status, environmental sensor data, and nearby BLE devices through an interactive 7-inch dashboard display [3], [5], [7].

LoRa communication enables reliable long-distance wireless sensor data transmission, while BLE supports nearby device scanning and smartwatch interaction. Wi-Fi is used for web server access, remote interaction, and Apple Shortcut battery updates [2], [6], [8]. The project also integrates a camera-based hand gesture control system using Edge Impulse machine learning for automatic screen rotation and user interaction [9]. A 7-inch display is connected to the main SmartHub device, where all real-time data and system information are visualized. A heavy-duty servo motor is used to rotate the display between 0° and 90° orientations for portrait and landscape viewing modes.

Overall, the SmartHub combines multiple IoT technologies into a single user-friendly dashboard platform designed to provide students and everyday users with convenient access to important real-time information from one centralized desk device.

Introduction

The Internet of Things (IoT) allows devices to connect, communicate, and share data automatically through the internet and wireless communication technologies. IoT systems improve convenience, automation, monitoring, and real-time interaction between smart devices. This project was developed to provide students with easier access to multiple smart functions through a single centralized dashboard system.

The IoT Connected SmartHub Dashboard is a multi-functional smart desk system developed using two Heltec WiFi LoRa ESP32 development boards and several additional ESP32-based components [1], [8]. The system integrates multiple communication technologies including LoRa for long-distance communication, Bluetooth Low Energy (BLE) for short-range device interaction, and Wi-Fi for web server access and remote communication [2], [6], [8].

The SmartHub supports real-time information such as weather updates, Spotify music playback, battery status, environmental sensor data, and nearby BLE device detection [5], [12]. The system also includes a 7-inch ESP32-based display used to visualize dashboard information and user interaction [3], [4], [7]. The display supports both portrait and landscape orientations through a

heavy-duty servo motor system [10]. With the help of camera-based gesture recognition, users can rotate the screen with simple hand gestures.

The overall goal of the project is to combine multiple IoT technologies into a single user-friendly SmartHub device designed for everyday student use.

Requirements/Specifications

Functional Requirements

- The SmartHub shall display real-time weather information.
- The system shall display Spotify music playback information.
- The system shall display environmental sensor data including temperature and humidity.
- The SmartHub shall support nearby BLE device scanning and detection.
- The system shall support long-distance LoRa communication between ESP32 nodes.
- The SmartHub shall provide Wi-Fi web server access for remote interaction and monitoring.
- The system shall support camera-based hand gesture recognition.
- The SmartHub shall support automatic screen rotation using a servo motor system.
- The dashboard shall support both portrait and landscape display orientations.
- The system shall support real-time updates for dashboard information and sensor values.

Non-Functional Requirements

- The system shall provide reliable wireless communication between subsystems.
- The dashboard interface shall remain user-friendly and responsive during operation.
- The LoRa transmitter node shall support low-power operation using deep sleep functionality.
- The system shall maintain stable operation during continuous runtime.
- The project shall support modular subsystem integration for easier debugging and maintenance.
- The system shall support scalable firmware development using PlatformIO environments.
- The SmartHub shall support smooth user interaction and display rendering using LVGL.
- The system shall maintain reliable servo motor operation without interrupting communication subsystems.

Research/Current Work review

The SmartHub Dashboard project started with individual subsystem testing to ensure that each component operated correctly and fulfilled its intended purpose. After completing subsystem-level testing, the project progressed toward full system integration using two Heltec ESP32 WiFi LoRa development boards and multiple wireless communication technologies. To complete the physical assembly of the system, a custom housing structure was implemented to support the main SmartHub ESP32 board, display stand, and camera holder. After completing the wiring and hardware integration, the entire system was tested to verify proper power distribution, subsystem communication, and overall functionality.

Current development focuses on integrating LoRa communication, BLE scanning, Wi-Fi web server functionality, environmental sensing, display rendering, gesture recognition, and power analysis for the overall SmartHub architecture.

The system currently uses a 7-inch Elecrow ESP32-S3 display running LVGL and LovyanGFX for the main dashboard interface [3], [4], [7]. BLE functionality has been implemented for nearby device scanning and smartwatch communication, while Wi-Fi supports web server access and Apple Shortcut battery updates for iPhone and iPad devices [6]. LoRa communication between ESP32 nodes has been successfully tested using 915 MHz SX1262 transceivers (TX and RX) for reliable long-distance environmental sensor data transmission [2], [8].

The project also integrates a camera-based hand gesture recognition subsystem using the Seeed Studio XIAO ESP32S3 Sense module and Edge Impulse machine learning [9]. The gesture recognition system currently supports directional gesture detection with approximately 92.9% accuracy.

Additional current work includes subsystem integration, automatic brightness control, power optimization, battery lifetime estimation, and final system debugging to improve overall system reliability and user experience

Design Method and overview

The SmartHub Dashboard was designed using a modular IoT architecture based on multiple ESP32 microcontrollers and wireless communication technologies [1]. The system integrates LoRa, Bluetooth Low Energy (BLE), and Wi-Fi communication into a centralized dashboard platform for real-time monitoring, interaction, and device control [2], [6], [8]. The very first architecture design is shown in Figure 2, while the final version of the architecture system for the smarthub is displayed in figure 1.

The main SmartHub ESP32 acts as the central controller and connects to multiple subsystems including the 7-inch Elecrow ESP32-S3 display, LoRa receiver node, environmental sensors, web server interface, speaker system, and camera-based gesture recognition module [7]. The dashboard interface was developed using LVGL and LovyanGFX to provide a responsive and user-friendly graphical display system [3], [4].

LoRa communication is implemented using Heltec ESP32 LoRa development boards operating at 915 MHz with SX1262 transceivers for reliable long-distance environmental data transmission [2], [8]. BLE functionality is used for nearby device scanning and smartwatch communication, while Wi-Fi provides web server access, remote interaction, and Apple Shortcut battery updates [6].

The system also integrates a camera-based gesture recognition subsystem using the Seeed Studio XIAO ESP32S3 Sense module and Edge Impulse machine learning [9]. Gesture detection is used for automatic screen rotation and dashboard interaction through directional hand gestures.

The overall design approach focuses on modular subsystem integration, reliable wireless communication, low-power operation for remote sensor nodes, and a user-friendly dashboard interface for everyday use.

(Include team member contribution and work completed)

Anar Amartuvshin

- Integrated BLE and Wi-Fi subsystems into the SmartHub dashboard architecture.
- Implemented nearby BLE device scanning and RSSI monitoring functionality.
- Developed Apple Shortcut battery update communication for iPhone and iPad devices through Wi-Fi.
- Designed and integrated the “Nearby Devices” dashboard interface.
- Assisted with camera dataset collection and gesture labeling for machine learning training.
- Designed the smartHub closure.

Johannes Hirt

- Integrated the LoRa receiver subsystem into the SmartHub architecture.
- Implemented LoRa packet reception, parsing, RSSI monitoring, and SNR measurement.
- Develop the SmartHub web server and remote-control interface.
- Integrated environmental sensing and automatic brightness control systems.

- Performed Packet Delivery Ratio (PDR) testing and received power analysis.
- Implemented and tested the servo motor subsystem.
- Assisted with camera dataset collection and gesture labeling.

Thiago Costa

- Developed the LVGL-based SmartHub dashboard interface for the 7-inch display.
- Implemented multiple dashboard pages including weather, battery, music playback, and BLE device visualization.
- Replaced the original OLED implementation with the larger touchscreen interface.
- Trained and optimized the Edge Impulse hand gesture recognition model.
- Improved camera gesture recognition performance and machine learning integration.
- Assisted with overall subsystem integration and UI improvements.

Overall Work Completed

- Full SmartHub dashboard integration using multiple ESP32 devices.
- Successful LoRa communication between ESP32 nodes at 915 MHz.
- BLE nearby device scanning and smartwatch interaction.
- Wi-Fi web server communication and battery update system.
- Camera-based hand gesture recognition with approximately 92.9% accuracy.
- Servo motor integration for automatic screen rotation.
- Environmental sensing and automatic brightness control.
- Power analysis and battery lifetime estimation for multiple subsystems.
- Real-time dashboard visualization using LVGL and LovyanGFX.

Verification and Result

The SmartHub Dashboard subsystems were verified through integration and communication testing across LoRa, BLE, Wi-Fi, display rendering, environmental sensing, and gesture recognition systems. Testing confirmed successful real-time communication between ESP32 nodes and stable dashboard operation.

LoRa communication testing demonstrated reliable packet transmission and reception at distances up to 100 meters under free-space conditions. Packet Delivery Ratio (PDR) measurements showed highly reliable communication performance, while received power measurements followed the expected logarithmic path loss behavior.

BLE testing confirmed successful nearby device scanning, RSSI monitoring, and smartwatch detection. Wi-Fi testing verified stable web server communication and successful Apple Shortcut battery updates for iPhone and iPad devices.

The display subsystem successfully rendered dashboard pages including weather information, battery status, environmental sensor values, and nearby BLE devices using the 7-inch ESP32-S3 display. Automatic brightness adjustment and portrait/landscape display modes were also successfully tested.

Gesture recognition testing using the ESP32-S3 camera module and Edge Impulse achieved approximately 92.9% accuracy for directional hand gestures including LEFT, RIGHT, UP, and DOWN.

Power analysis results confirmed stable subsystem operation and demonstrated the effectiveness of separating the system into multiple isolated power domains for improved reliability and reduced power-related communication interruptions.

Power and lifetime analysis/measurements

To determine the realistic power requirements and to estimate the battery lifetime for each subsystem, we have performed the power analysis where it is needed. Measurements were collected for the SmartHub receiver, LoRa transmitter node, and servo motor system. The SmartHub receiver node, which includes the 7-inch display, Wi-Fi connection, and web server, was measured to consume approximately 470 mA continuously during operation. Due to the high current draw of the display and continuous Wi-Fi activity, we have decided to operate using wall power through a dedicated 5V power supply for the receiver system. Another alternativity is that we may use a large battery and use the voltage regulator where we got from the parts room. That way can find the way to supply 5V to the ESP32 dev boards. Since the servo motor draws large current during the rotation, we have analyzed separately from the main system. Our measurements showed current peaks up to 8.2 A with an average active current of approximately 2.44 A during rotation. To prevent voltage drops and ESP32 brownout resets, the servo motor is powered using an independent battery source isolated from the main SmartHub controller.

The remote LoRa environmental sensor node consumes significantly low power due to deep sleep operation between transmissions, which acts like a low power mode. The transmitter node consumes approximately 129 mA during LoRa transmission and approximately 0.5 mA during idle or deep sleep states. The average current consumption for the LoRa transmitter node was estimated using the following relationship: Using a 5-minute transmission interval and a 2-second active transmission duration, the estimated average current consumption was

approximately 1.356 mA. Battery lifetime analysis was also performed using a 2000 mAh battery with an 85% usable capacity factor. Based on the measured values:

- The remote LoRa transmitter node is estimated to operate for approximately 52 days.
- The isolated servo battery system is estimated to operate for approximately 20 days under periodic sweep operation.
- The SmartHub receiver node would operate for only several hours on battery power and therefore remains wall powered.

The power analysis validates the decision to separate the system into multiple isolated power domains. This architecture improves reliability and prevents high-current motor events from interrupting wireless communication and dashboard operation.

Conclusions

The IoT Connected SmartHub Dashboard successfully demonstrates the integration of multiple IoT technologies into a single user-friendly smart desk system. The project combines LoRa, Bluetooth Low Energy (BLE), and Wi-Fi communication with environmental sensing, web server functionality, graphical dashboard rendering, and camera-based gesture recognition using ESP32 microcontrollers.

Testing results demonstrated reliable LoRa communication, successful BLE device scanning, stable Wi-Fi interaction, and accurate hand gesture recognition performance. Power analysis and battery lifetime measurements also validated the use of isolated power domains to improve system stability, communication reliability, and power efficiency.

Overall, the SmartHub Dashboard provides students and everyday users with convenient access to useful real-time information through a centralized and interactive platform while demonstrating the practical implementation of modern IoT communication, embedded systems, and edge machine learning technologies.

Recommendations

- Improve final subsystem integration and reduce communication delays between ESP32 devices.
- Continue optimizing the hand gesture recognition model to reduce false detections and improve overall accuracy.
- Improve power management and battery efficiency for more portable system operation.

- Add additional environmental sensors to expand monitoring capabilities.
- Improve the SmartHub user interface with smoother animations and improved screen organization.
- Expand LoRa communication testing in more realistic indoor and outdoor environments.
- Implement more secure communication methods for Wi-Fi and BLE connections.
- Improve hardware organization using a custom 3D-printed enclosure and mounting system.
- Add cloud connectivity for remote monitoring, synchronization, and data storage.
- Continue long-term reliability and stress testing under continuous system operation.

References

[1] Espressif Systems, *ESP32 Technical Reference Manual*, Espressif Systems, 2024.

[2] Semtech Corporation, *SX1261/2 Data Sheet*, Semtech Corporation, 2023.

[3] [LVGL Graphics Library Documentation](#)

[4] [LovyanGFX Graphics Library for ESP32](#)

[5] [OpenWeatherMap Current Weather Data API](#)

[6] [Arduino Documentation – WiFi Web Server Example](#)

[7] [Elecrow ESP32-S3 7-inch Display Documentation](#)

[8] [Heltec WiFi LoRa 32 Documentation](#)

[9] [Edge Impulse Documentation](#)

[10] [ESP32Servo Library](#)

[11] [football-data.org API Documentation](#)

[12] [Spotify Web API Documentation](#)

Appendices

A System Architecture

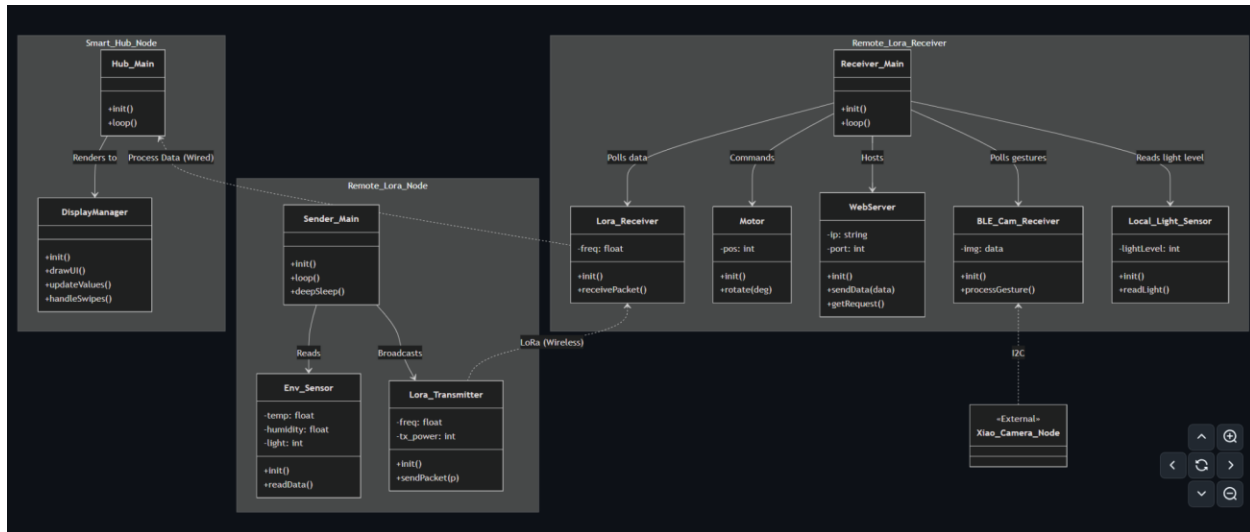


Figure 1: Overall smartHub Architecture

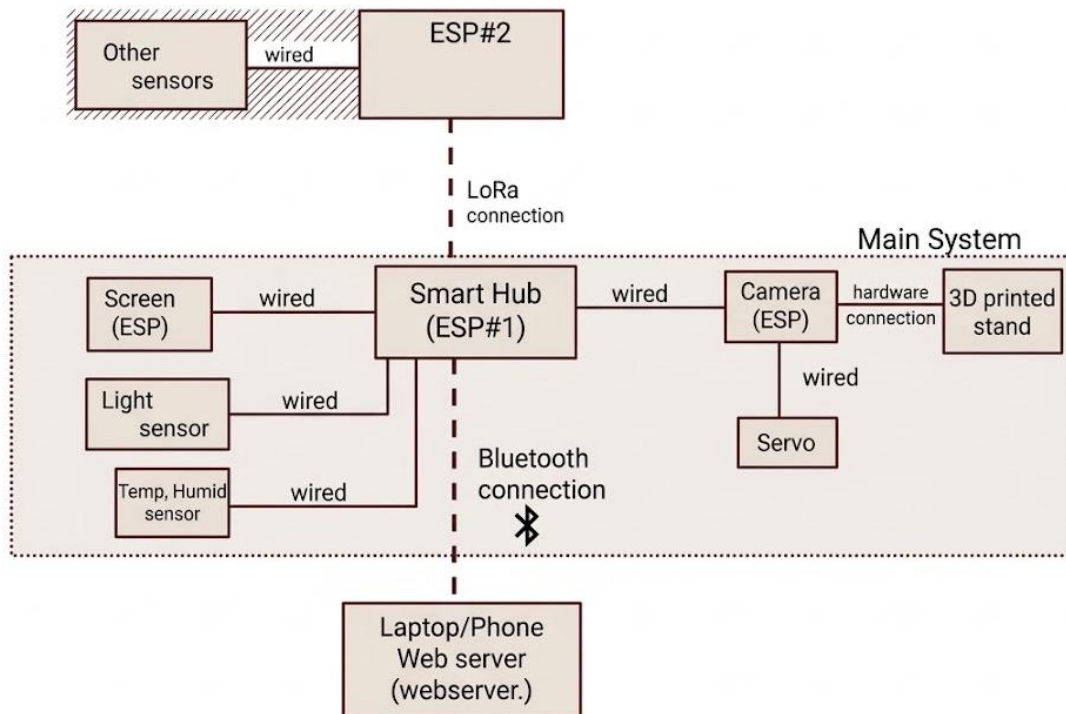


Figure 2: The first architecture design for the overall system

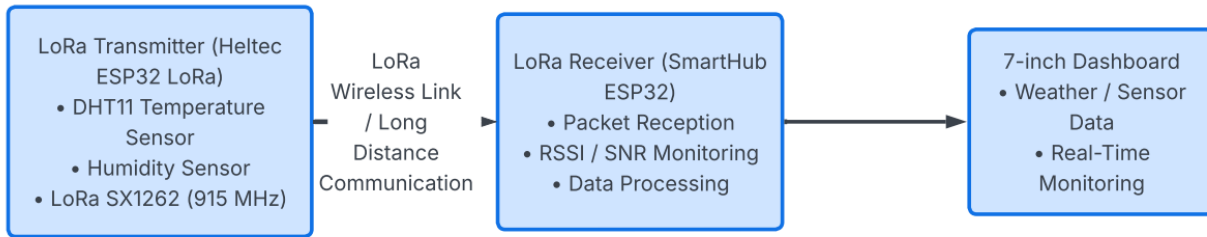


Figure 3: LoRa communication block diagram

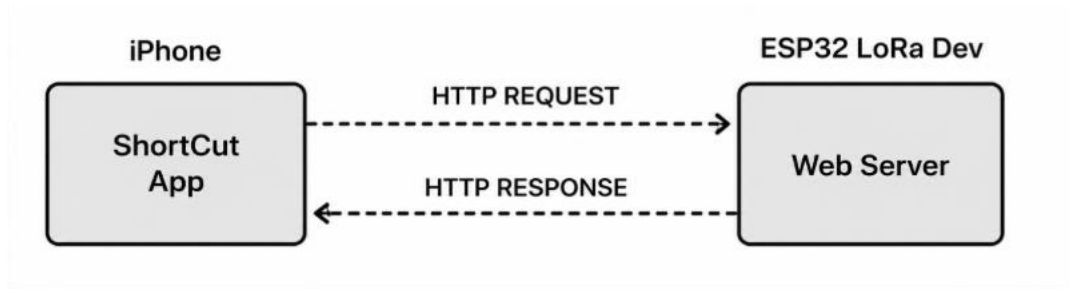


Figure 4: Wi-Fi and BLE architecture diagram

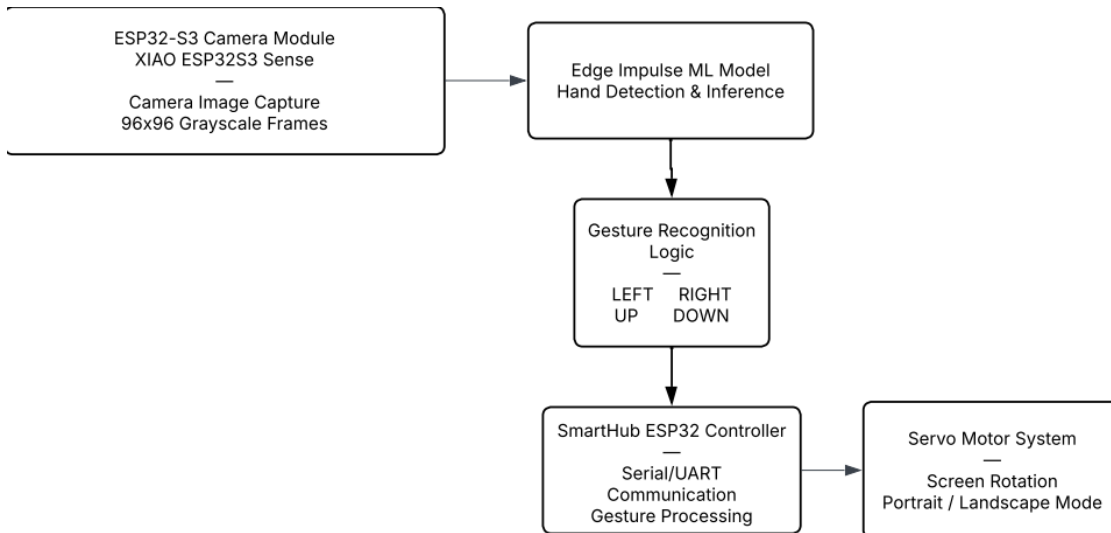
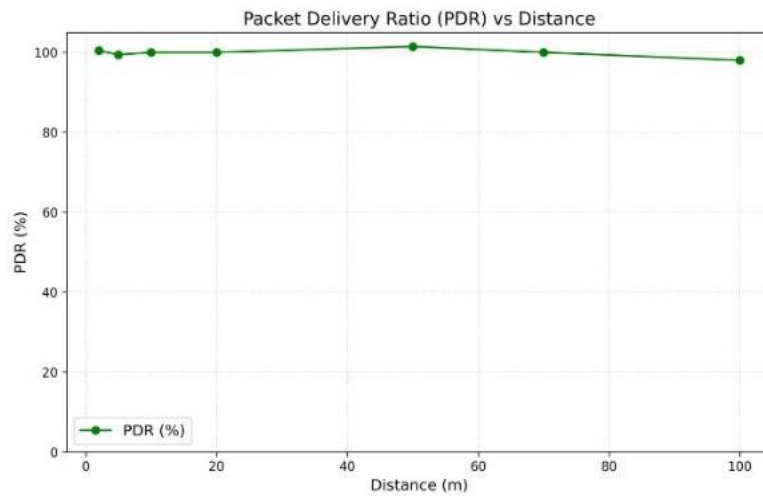
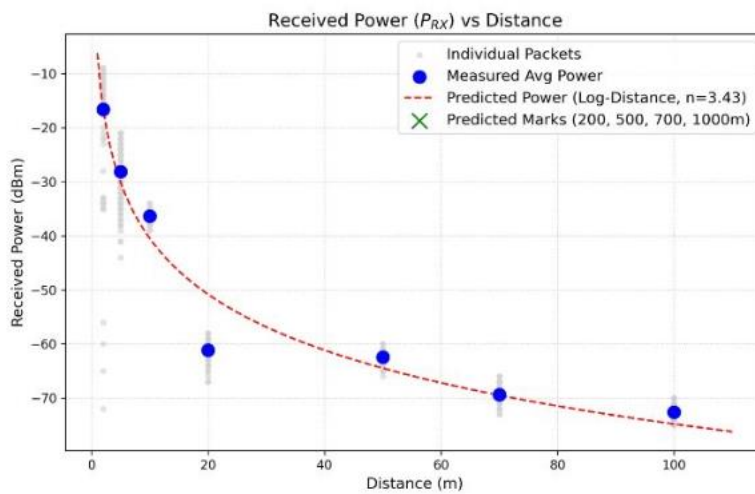


Figure 5: Camera gesture system architecture

B Circuit Diagrams



Testing 1: Packets Delivery ration (PDR) table



Testing 2: Received power vs Distance

Testing 3: RSSI measurements

Testing 4: Power measurements

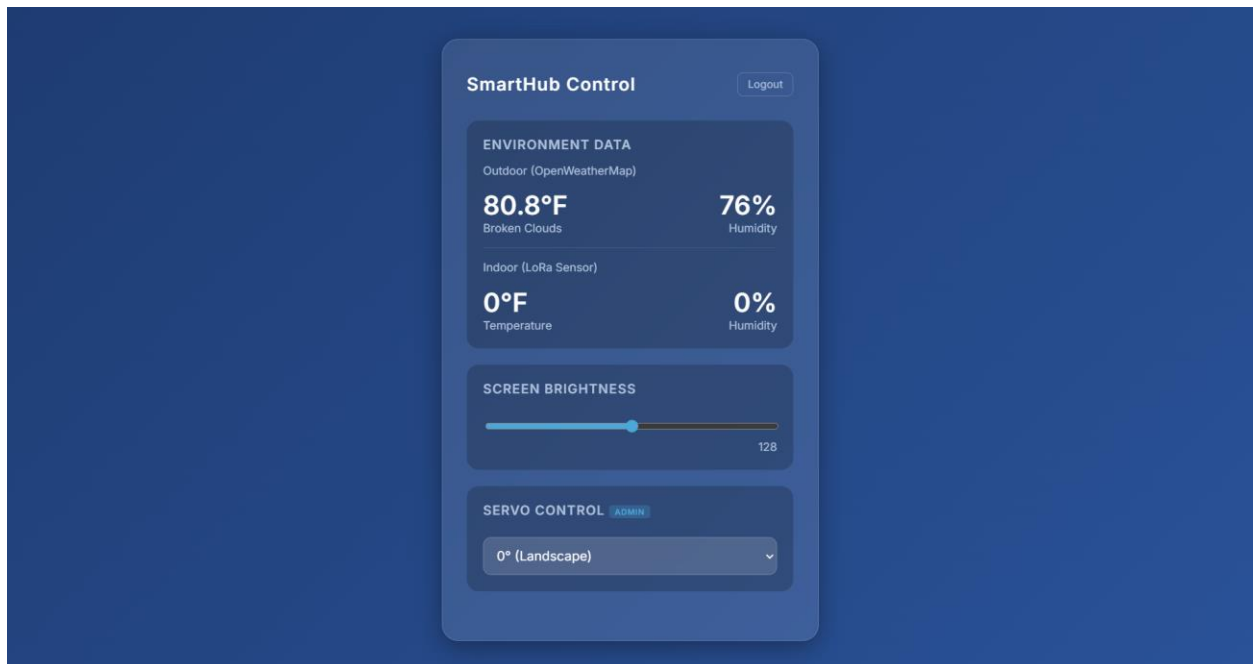
Testing 5: Gesture accuracy results

Testing 6: Battery lifetime calculations

E Ui screenshots

Screenshot 1: Main dashboard

Screenshot 2: BLE page



Screenshot 3: Weather screen

Screenshot 4: Web server interface



Screenshot 5: Gesture recognition demo

F Power Analysis calculations

Figure 1: Current measurement calculations

Figure 2: Average current formulas

Figure 3: Battery lifetime equations

Figure 4: Deep sleep calculations

G Bill of Materials

Items	Details	Costs \$
Screen	7-inch 800x480 ESP32 Screen	53.49 \$
Camera	Xiao ESP32 S3 Camera	21.83 \$
Motor	20kg RC servo High Torque Servo Motors	14 \$
Photoresistor	We have it	
Humidity sensor	We have it	
Temperature Sensor	We have it	
total		89.32 \$
Cost per person		29.73 \$

Table 1: Bill of materials

Appendix H- team contributions

Table 1: team contribution table